



CVL2001: Climate Change and Adaptation

Lecture 4: Impacts of Climate Change

3.2 Ocean Acidification, Hurricanes and Extreme Event Attribution

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Section B

Quick Recap

Impacts of Climate Change

1. Temperature
2. Precipitation
3. Sea Level Rise
4. Ocean Acidification
5. Hurricanes/Cyclones

4. Ocean Acidification

The ocean absorbs part of the CO₂ released into the atmosphere.

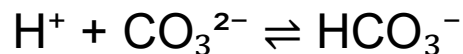
1. Dissolved CO₂ reacts with water to form carbonic acid:



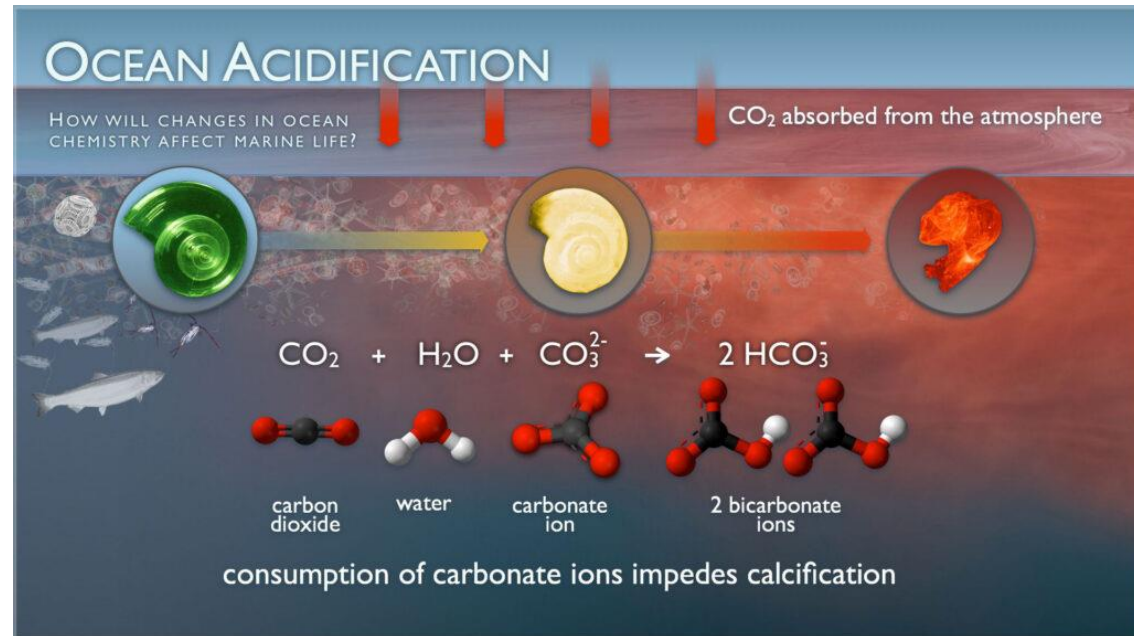
2. Carbonic acid releases hydrogen ions, which lower pH:



3. The additional hydrogen ions react with carbonate already present in seawater, converting it into bicarbonate:



Seawater remains alkaline, but its pH is falling and less carbonate is available for shell and skeleton formation.



Basic chemistry of ocean acidification and its effects on shelled organism. Credit: NOAA Pacific Marine Environmental Laboratory

4. Ocean Acidification

- The pH scale is logarithmic, so a small decrease in pH represents a substantial increase in acidity.
- Ocean acidification reduces the carbonate ions needed by many organisms to build shells and skeletons.
- Corals, shellfish and marine food webs must adapt to a rapidly changing chemical environment.

Q) Who—and what—is at risk when marine ecosystems change?



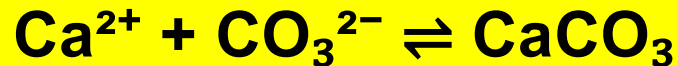
Coral Bleaching. Before and after images of the same reef showing the effects of coral bleaching in American Samoa.

Image source: <https://www.climate.gov/news-features/event-tracker/survey-photos-reveal-damage-year%E2%80%99s-global-coral-bleaching-event>

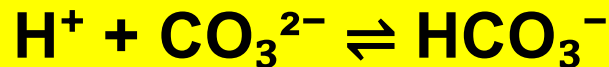
EXTRA NOTE

Q) Why Does Carbonate Decrease?

- Carbonate is naturally **used and recycled** in the ocean.
- Organisms use it to build shells and skeletons:



- Some calcium carbonate later dissolves, while rivers and sediments help replenish the ocean's carbonate system.
- Ocean acidification introduces an additional change:



- Extra CO_2 produces more H^+ , which converts existing carbonate into bicarbonate.
- **Without added CO_2 :** carbonate continues through its natural cycle.
With added CO_2 : bicarbonate increases and carbonate availability decreases.

Result: less carbonate is available for corals and shell-forming organisms.

Source: <https://oceanacidification.noaa.gov/what-is-ocean-acidification/>

5. Hurricanes

- A hurricane is an intense rotating tropical cyclone that forms over warm ocean water.
- The same type of storm is called a cyclone in the Indian Ocean and a typhoon in the northwestern Pacific.
 - Warmer oceans can support stronger storms.
 - A warmer atmosphere can increase rainfall within storms.
 - Sea-level rise allows storm surges to reach farther inland.

The main concern is not simply more storms, but more damaging ones.



<https://earthsky.org/up/2022/09/storm-cloud-crashing-wave-pexels-george-desipris.jpeg>

5. Hurricanes...

What Is Changing About Tropical Cyclones?

- **Frequency:** The total number of tropical cyclones is projected to decrease or remain broadly unchanged globally.
- **Intensity:** Average peak wind speeds and the proportion of very intense storms are projected to increase.
- **Rainfall:** Tropical cyclones are expected to produce heavier rainfall as the atmosphere warms.
- **Storm surge:** Sea-level rise increases the height and inland reach of storm-surge flooding.

Climate change may not produce more tropical cyclones, but the storms that occur can become more damaging.

Quick Recap: Impacts of Climate Change

1. **Temperature anomaly** — Global and regional temperatures are rising.
2. **Precipitation anomaly** — Rainfall patterns and extremes are changing.
3. **Sea-level rise** — Warming oceans and melting land ice raise sea level.
4. **Ocean acidification** — Absorbed CO₂ lowers ocean pH and reduces carbonate availability.
5. **Tropical cyclones/hurricanes** — The strongest storms can become more intense, wetter and more damaging.

Next: Can climate change influence an individual extreme event?

Extreme Event Attribution

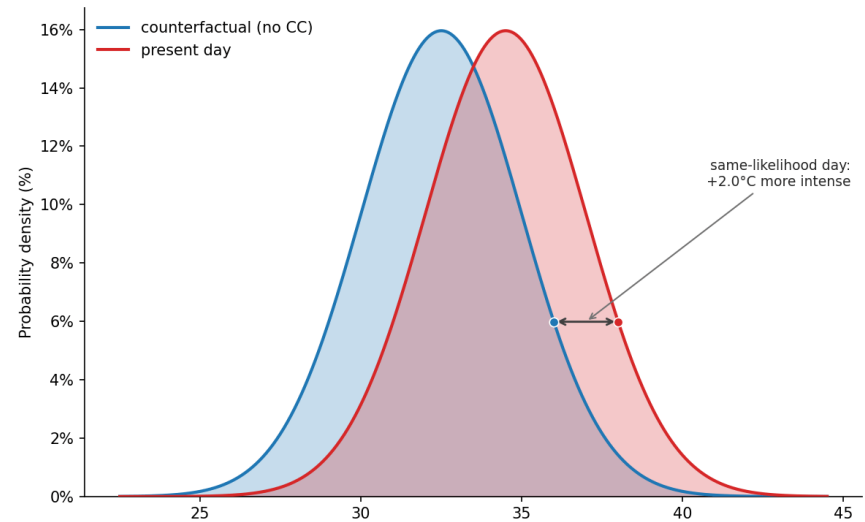
- Estimates the influence of human-caused climate change on a specific extreme event.
- Measures changes in:
 - ✓ **Probability:** Did the event become more likely?
 - ✓ **Intensity:** Did the event become stronger?

Attribution does not identify a single cause; it measures how climate change influenced the event.

Actual vs Counterfactual

- **Actual world**
Observations and models include human and natural influences.
- **Counterfactual world**
Models include natural influences but remove human-caused warming.

The difference between the two worlds represents the estimated human influence.



Example comparison Image Source:

[https://substackcdn.com/image/fetch/\\$s!pzo7!,w_1456,c_limit,f_auto,q_auto:good,fl_progressive:steep/https%3A%2F%2Fsubstack-post-media.s3.amazonaws.com%2Fpublic%2Fimages%2Fefb0c3ba-809d-43fb-9bd0-c6a16723d2c2_1200x750.png](https://substackcdn.com/image/fetch/$s!pzo7!,w_1456,c_limit,f_auto,q_auto:good,fl_progressive:steep/https%3A%2F%2Fsubstack-post-media.s3.amazonaws.com%2Fpublic%2Fimages%2Fefb0c3ba-809d-43fb-9bd0-c6a16723d2c2_1200x750.png)

Case Study: Wayanad Landslide, 30 July 2024



1. Yunus, A. P., Sajinkumar, K. S., Gopinath, G., Siva Subramanian, S., Kaushal, S., Thanveer, J., Achu, A. L., Ul Islam, S. M., Ishan, A., Krishnapriya, V. K., Rajaneesh, A., Dewrari, M., Dixit, S., Singh, S., Srivastava, P., Oommen, T., Nedumpallile-Vasu, N., Sen, S., Narayana, A. C., Ambili, V., Pradeep, G. S., & Kuriakose, S. L. (2025). Chronicle of destruction: The Wayanad landslide of July 30, 2024. *Landslides*, 22, 1891–1908. <https://doi.org/10.1007/s10346-025-02494-y>
2. Das, R. (2025). Catastrophic landslide in Wayanad district of Kerala, India on July 30, 2024: A complex interplay between geology, geomorphology, and climate. *Landslides*, 22, 271–281. <https://doi.org/10.1007/s10346-024-02385-8>

Case Study: Wayanad Landslide, 30 July 2024

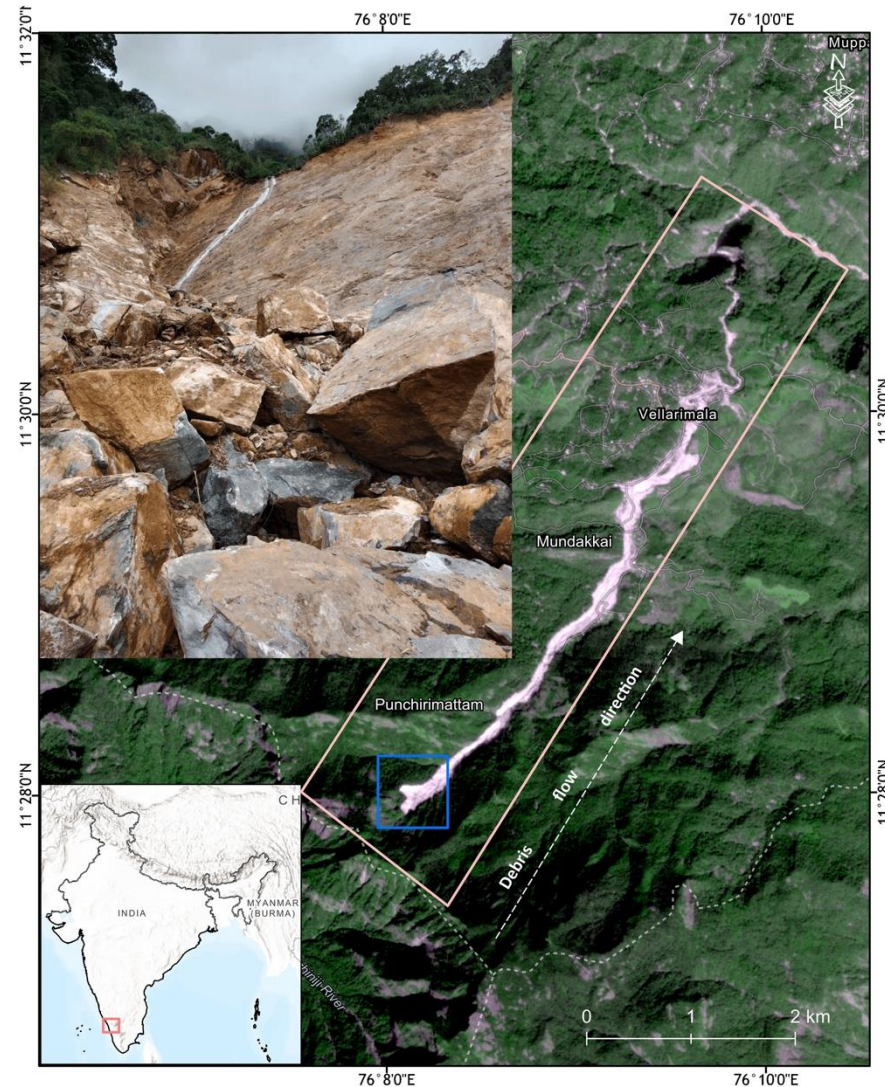
- A catastrophic landslide began in the hills above Meppadi, Wayanad (Kerala).
- The material transformed into a fast-moving debris flow and travelled approximately 8 km downstream.
- Mundakkai, Chooralmala and Punchirimattom were severely affected.

Investigation Question: Was this disaster influenced by climate change—or was it primarily the result of rainfall, geology and local slope conditions?

What Happened?

- On 30 July 2024, a slope failed in the hills above Meppadi, Wayanad.
- Failure began during the early morning, around 1–2 a.m.
- The landslide transformed into a fast-moving debris flow.
- Debris travelled approximately 8 km along the river valley.
- Punchirimattam, Mundakkai and Chooralmala were severely affected.

Investigation question: What transformed a slope failure into a catastrophic disaster?



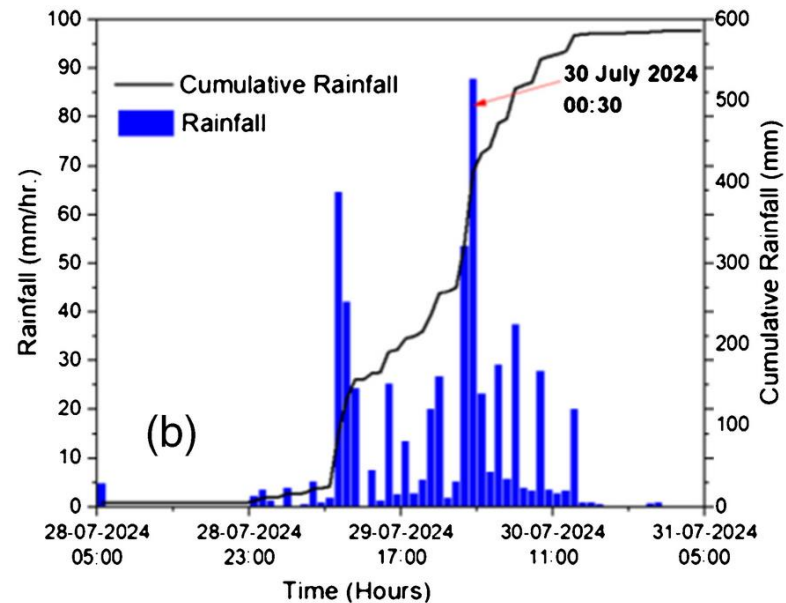
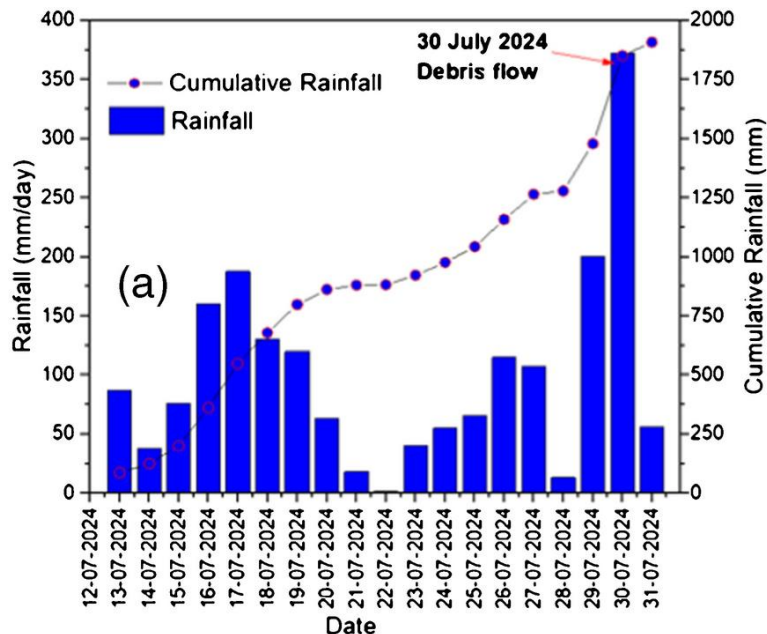
The main frame shows the satellite view of the Wayanad debris flow (inside light orange box). Landslide scrap is marked by blue box, and a field photograph of the same is shown above. (Yunus et al., 2025)

What Does the Rainfall Data Show?

- The landslide followed an exceptionally intense monsoon spell.
- 586 mm of rainfall was recorded over 29–30 July.
- Prolonged rainfall saturated the soil and increased pore-water pressure.
- Extreme rainfall was the immediate trigger.

Question: Was extreme rainfall sufficient to explain the disaster?

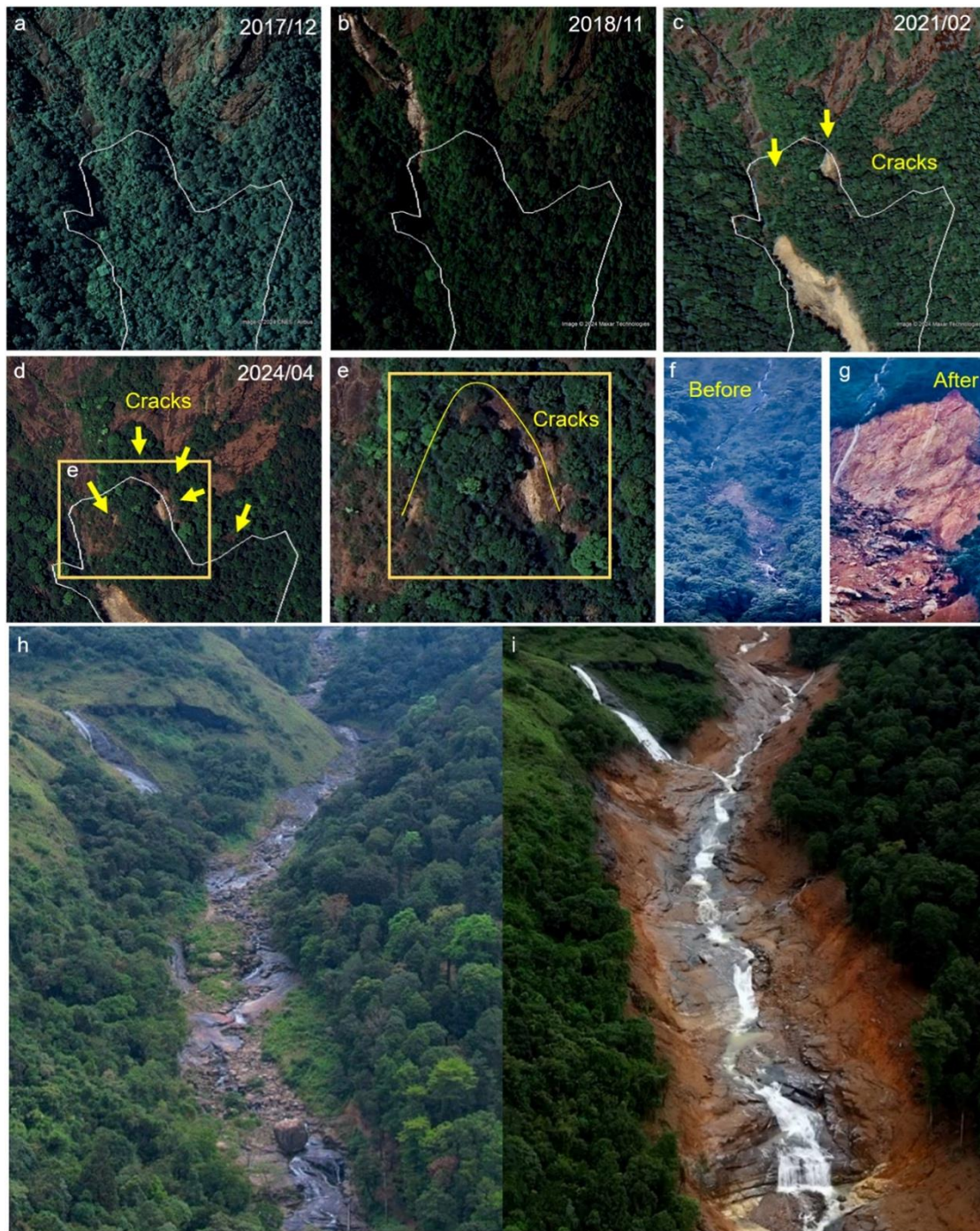
Daily rainfall and **b** hourly rainfall data.
Yunus et al.
2025



What Made the Slope Susceptible?

- Rainfall acted on a slope that was already unstable.
- A pre-existing crack had developed during a 2020 landslide.
- The slope contained highly weathered and fractured rock.
- Thick, loose soil covered the steep terrain.
- Water infiltration reduced the strength of the slope material.
- The narrow river channel concentrated the debris flow downstream.

Rainfall triggered the failure; slope conditions controlled how it occurred.



Source-area evolution: (a)

Undisturbed slope, 2017;

(b) minor debris flow after the 2018 monsoon;

(c) 2020 debris flow and newly developed cracks;

(d-e) widening cracks before the 2024 event;

(f, h) source and transition zones before failure;

(g, i) after the 30 July 2024 debris flow.

What Did the Two Studies Conclude?

Yunus et al. (2025)

A geologically susceptible slope failed during intense monsoon rainfall and transformed into a destructive debris flow.

Das (2025)

The event reflected an interaction between geology, geomorphology, extreme rainfall, climate and local environmental conditions.

What Role Could Climate Change Have Played?

- Human-caused warming may have intensified the rainfall hazard.
- Attribution research estimated that warming made the one-day rainfall approximately **10% heavier**.
- This does not mean climate change directly caused the slope to fail.
- Geology, antecedent rainfall and slope conditions influenced the landslide.
- Runout, exposure and vulnerability determined the scale of the disaster.
- **Climate change → heavier rainfall → slope failure → debris flow → disaster**

What Does the Evidence Mean?

Complete the explanation:

1. Climate change influenced

2. The slope failed because

3. The event became a disaster because

Public Debate: What Caused the Disaster?

“Climate change caused it.”

“Deforestation caused it.”

“Extreme rainfall caused it.”

“Poor planning caused it.”



These claims compete only if we expect a single cause.

A disaster is not one event with one cause; it is a chain of interacting causes.

So, Did Climate Change Cause the Disaster?

- Climate change intensified the rainfall hazard.
- Slope conditions determined whether the slope failed.
- Exposure and vulnerability determined the consequences.

Climate change contributed—but it was not the sole cause.

Q) Can 10% More Rain Make a Large Difference?

- A slope can absorb rainfall until it reaches a critical limit.
- A small increase may then trigger a much larger response:

10% more rainfall → threshold crossed → slope failure

Next: Nonlinear Impacts and Thresholds